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Liquid ejecting system

Abstract

A liquid ejecting system includes a head unit that ejects liquid, a first output portion that outputs unique information unique to the head unit to an external server, a first input portion to which update information about the head unit is input from the external server in association with the unique information, and a notification portion that notifies a user based on the update information.

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2021-030689	12/2020	JP	N/A

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Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2022-088364, filed May 31, 2022, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to a liquid ejecting system.

2. Related Art

(3) In a liquid ejecting apparatus such as an ink jet printer, generally, liquid such as ink is ejected from a liquid ejecting head by supplying a drive pulse to a drive element such as a piezoelectric element. For example, JP-A-2021-030689 discloses a liquid ejecting apparatus including a plurality of heads.

(4) In recent years, there has been a business model in which a head manufacturer sells only a head to a printer manufacturer. Here, the head manufacturer may want to update the head in some way after shipping the head. For example, when some kind of defect is found for the first time after the head is shipped, there is a request to notify users of the printer sold by the printer manufacturer as soon as possible. Further, even when the program incorporated in the head is improved after the head is shipped, there is also a request that the users want to update the program as soon as possible. However, in the sales of the head, the printer manufacturer and the like intervenes between the head manufacturer and the user, and thus it is likely that the head manufacturer is not able to specify the user, and as a result, is unable to immediately inform the user about information on updates as described above.

SUMMARY

(5) According to an aspect of the present disclosure, there is provided a liquid ejecting system including a head unit that ejects liquid, a first output portion that outputs unique information unique to the head unit to an external server, a first input portion to which update information about the

head unit is input from the external server in association with the unique information, and a notification portion that notifies a user based on the update information.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a schematic diagram showing a configuration example of a liquid ejecting system according to a first embodiment.

(2) FIG. **2** is a schematic diagram showing a configuration example of a liquid ejecting apparatus used in the liquid ejecting system according to the first embodiment.

(3) FIG. **3** is a cross-sectional view showing a configuration example of a head chip.

(4) FIG. **4** is a schematic diagram showing a configuration example of a first processing apparatus used in the liquid ejecting system according to the first embodiment.

(5) FIG. **5** is a schematic diagram showing a configuration example of an external server used in the liquid ejecting system according to the first embodiment.

(6) FIG. **6** is a flowchart showing a process of the liquid ejecting system according to the first embodiment.

(7) FIG. **7** is a diagram showing Arrangement Example 1 of a plurality of liquid ejecting heads.

(8) FIG. **8** is a diagram showing an example in which an arrangement image showing Arrangement Example 1 is displayed in a display region.

(9) FIG. **9** is a diagram showing an example in which a property image and an input image are displayed in the display region.

(10) FIG. **10** is a diagram showing Arrangement Example 2 of a plurality of liquid ejecting heads.

(11) FIG. **11** is a diagram showing an example in which an arrangement image showing Arrangement Example 2 is displayed in the display region.

(12) FIG. **12** is a diagram showing Arrangement Example 3 of a plurality of liquid ejecting heads.

(13) FIG. **13** is a diagram showing an example in which an arrangement image showing Arrangement Example 3 is displayed in the display region.

(14) FIG. **14** is a diagram showing Arrangement Example 4 of a plurality of liquid ejecting heads.

(15) FIG. **15** is a diagram showing an example in which an arrangement image showing Arrangement Example 4 is displayed in the display region.

(16) FIG. **16** is a schematic diagram showing a configuration example of a first processing apparatus according to a second embodiment.

(17) FIG. **17** is a schematic diagram showing a configuration example of a liquid ejecting system according to a third embodiment.

(18) FIG. **18** is a schematic diagram showing a configuration example of a liquid ejecting apparatus used in the liquid ejecting system according to the third embodiment.

(19) FIG. **19** is a flowchart showing a process of the liquid ejecting system according to the third embodiment.

(20) FIG. **20** is a schematic diagram showing a configuration example of a liquid ejecting system according to a fourth embodiment.

(21) FIG. **21** is a flowchart showing a process of the liquid ejecting system according to the fourth embodiment.

DESCRIPTION OF EXEMPLARY EMBODIMENTS

(22) Hereinafter, preferred embodiments according to the present disclosure will be described with reference to the accompanying drawings. In the drawings, the dimensions and scale of each portion are appropriately different from the actual ones, and some portions are schematically shown for easy understanding. Further, the scope of the present disclosure is not limited to the embodiments unless it is stated in the following description that the present disclosure is particularly limited.

1. First Embodiment

(23) 1-1. Outline of Liquid Ejecting System

(24) FIG. **1** is a schematic diagram showing a configuration example of a liquid ejecting system **10** according to a first embodiment. The liquid ejecting system **10** is a system that performs printing by an ink jet method. In the example shown in FIG. **1**, the liquid ejecting system **10** includes liquid ejecting apparatuses **100_1** to **100_3**, first processing apparatuses **200_1** to **200_3**, an external server **300**, and a second processing apparatus **400**.

(25) Here, each of the liquid ejecting apparatuses **100_1** to **100_3** is provided by a printer manufacturer that manufactures printer main bodies, which will be described later. The liquid ejecting apparatuses **100_1** to **100_3** may be provided by the same manufacturer or may be provided by different manufacturers. Each of first processing apparatuses **200_1** to **200_3** may use a property of a user or may be provided by a printer manufacturer. On the other hand, a head unit **110** incorporated in each of the liquid ejecting apparatuses **100_1** to **100_3** is provided by a head manufacturer that manufactures heads, which will be described later. The external server **300** may be any server as long as the head manufacturer can provide the necessary services to the user, and may be owned by the head manufacturer itself, or by a third party other than the head manufacturer. The second processing apparatus **400** is owned by the head manufacturer itself. The second processing apparatus **400** is maintained and managed by the head manufacturer.

(26) For example, the user using the liquid ejecting apparatus **100_1** owns the liquid ejecting apparatus **100_1**, the first processing apparatus **200_1**, and the head unit **110**. On the other hand, although the user does not own the external server **300**, the first processing apparatus **200_1** owned by the user is communicably connected to the external server **300** through a communication network NW. When the external server **300** is owned by a third party, the second processing apparatus **400** is communicably connected to the external server **300** through the communication network NW (not shown).

(27) For example, when a printer manufacturer for manufacturing a printer main body using a head purchased from the head manufacturer uses the printer main body manufactured by itself, the printer manufacturer is the user. Further, for example, when a printer manufacturer that manufactures a printer main body using a head purchased from the head manufacturer sells the printer main body to a third party and the third party uses the printer main body, the third party is the user.

(28) The liquid ejecting apparatus **100_1** is communicably connected to the first processing apparatus **200_1**. The liquid ejecting apparatus **100_2** is communicably connected to the first processing apparatus **200_2**. The liquid ejecting apparatus **100_3** is communicably connected to the first processing apparatus **200_3**. As described above, the liquid ejecting apparatuses **100_1** to **100_3** correspond to the first processing apparatuses **200_1** to **200_3**, respectively, and are communicably connected to the first processing apparatuses **200_1** to **200_3**. In the following, without distinguishing each of the liquid ejecting apparatuses **100_1** to **100_3** from the others, they may be referred to as the liquid ejecting apparatus **100**. Without distinguishing each of the first processing apparatuses **200_1** to **200_3** from the others, they may be referred to as the first processing apparatus **200**.

(29) In the example shown in FIG. **1**, the number of each of the liquid ejecting apparatus **100** and the first processing apparatus **200** included in the liquid ejecting system **10** is three, but the number is not limited thereto, and the number may be one, two, or four or more. That is, the number of sets of the liquid ejecting apparatus **100** and the first processing apparatus **200** is not limited to three, and may be one, two, or four or more.

(30) The liquid ejecting apparatus **100** is a printer that prints an image based on recorded data DP from the first processing apparatus **200** on a print medium by an ink jet method. The recorded data DP is image data in a format that can be processed by the liquid ejecting apparatus **100**. The print medium may be any medium as long as it can be printed by the liquid ejecting apparatus **100**, and

is not particularly limited, and is, for example, various papers, various cloths, various films, and the like. The liquid ejecting apparatus **100** may be a serial type printer or a line type printer.
(31) The liquid ejecting apparatus **100** has a plurality of head units **110**. The head units **110** are modules including an ink jet head. In the following, among the elements constituting the liquid ejecting apparatus **100**, the elements other than the head unit **110** may be referred to as a “printer main body”. Further, the head unit **110** or a liquid ejecting head **110***a* to be described later may be simply referred to as a “head”. A configuration of the liquid ejecting apparatus **100** will be described in detail later with reference to FIGS. **2** and **3**.
(32) The first processing apparatus **200** is a desktop-type, a laptop-type computer, or the like, having a function as a management apparatus for managing the liquid ejecting apparatus **100**. In addition to the function, the first processing apparatus **200** has a function of generating recorded data DP and a function of controlling printing by the liquid ejecting apparatus **100**. A configuration of the first processing apparatus **200** will be described in detail later with reference to FIG. **4**.
(33) The first processing apparatus **200** is communicably connected to the external server **300** through the communication network NW including the Internet. The first processing apparatus **200** has a function of outputting unique information D**1** to the external server **300**, a function of inputting update information D**2** from the external server **300**, a function of performing notification based on the update information D**2**, and a function of displaying management information the liquid ejecting apparatus **100**. The unique information D**1** is information unique to the head unit **110** or the liquid ejecting head **110***a*, and is, for example, identification information indicating a unique number such as a serial number. The update information D**2** is information on updating the head unit **110** or the liquid ejecting head **110***a*, and is, for example, information for notifying a deficiency, information for updating a program, such as firmware for ejecting ink, and the like. Further, the first processing apparatus **200** generates the recorded data DP by performing various processing, such as raster image processor (RIP) processing or color conversion processing, on image data in the file format such as PostScript, a portable document format (PDF), and an XML paper specification (XPS).
(34) The external server **300** is a computer that functions as a cloud server, and has a function of inputting the unique information D**1** from the first processing apparatus **200** and a function of outputting the update information D**2** according to the unique information D**1** to the first processing apparatus **200**. A configuration of the external server **300** will be described in detail later with reference to FIG. **5**.
(35) Further, the external server **300** is communicably connected to the second processing apparatus **400**, and appropriately transmits and receives information necessary for providing the update information D**2** to the user. The second processing apparatus **400** is a computer that outputs, to the external server **300**, information necessary for providing the update information D**2** to the user.
(36) In the liquid ejecting system **10** outlined above, the first processing apparatus **200** outputs the unique information D**1** to the external server **300**, and thus the external server **300** can generate the update information D**2** based on the unique information D**1**. Then, the first processing apparatus **200** performs the notification based on the update information D**2** input from the external server **300**, and thus, the update information D**2** is provided to the user. As described above, even when the printer manufacturer is different from the head manufacturer, the update information D**2** about the head unit **110** can be promptly provided to the user. Hereinafter, the liquid ejecting system **10** will be described in detail.
(37) 1-2. Configuration of Liquid Ejecting Apparatus
(38) FIG. **2** is a schematic diagram showing a configuration example of the liquid ejecting apparatus **100** used in the liquid ejecting system **10** according to the first embodiment. As shown in FIG. **2**, the liquid ejecting apparatus **100** includes a plurality of head units **110**, a moving mechanism **120**, a communication device **130**, a storage circuit **140**, and a processing circuit **150**.

(39) The head unit **110** is an assembly including a head chip **111**, a drive circuit **112**, a power supply circuit **113**, a drive signal generation circuit **114**, a storage circuit **116**, and a processing circuit **117**. The storage circuit **116** is an example of a “storage portion”.
(40) In the example shown in FIG. **2**, the head unit **110** is divided into the liquid ejecting head **110***a* including the head chip **111** and the drive circuit **112**, and a control module **110***b* including the power supply circuit **113**, the drive signal generation circuit **114**, the storage circuit **116**, and the processing circuit **117**. The head unit **110** is not limited to the aspect of being divided into the liquid ejecting head **110***a* and the control module **110***b*, and for example, a part or all of the control module **110***b* may be incorporated in the liquid ejecting head **110***a*.
(41) In FIG. **2**, one liquid ejecting head **110***a* is typically illustrated for convenience of drawing, but the head unit **110** has a plurality of liquid ejecting heads **110***a*. An example of arranging the plurality of liquid ejecting heads **110***a* will be described later with reference to FIGS. **7**, **10**, **12**, and **14**.
(42) The head chip **111** ejects ink toward the print medium. In FIG. **2**, among the components of the head chip **111**, a plurality of drive elements **111***f* are typically shown. A detailed example of the head chip **111** will be described later with reference to FIG. **3**.
(43) In the example shown in FIG. **2**, the head unit **110** has one head chip **111** in number, but the number may be two or more. When the liquid ejecting apparatus **100** is a serial type, one or more head chips **111** are arranged so that a plurality of nozzles are distributed over a part of the width direction of the print medium. Further, when the liquid ejecting apparatus **100** is a line type, two or more head chips **111** are arranged so that a plurality of nozzles are distributed over the entire width direction of the print medium.
(44) The drive circuit **112** performs switching under the control of the processing circuit **150** as to whether or not to supply a drive signal Com output from the drive signal generation circuit **114** to each of the plurality of drive elements **111***f* of the head chip **111** as a drive pulse PD. The drive circuit **112** includes, for example, a group of switches such as a transmission gate for the switching.
(45) The power supply circuit **113** receives electric power from a commercial power source (not shown) and generates various predetermined potentials. The various potentials generated are appropriately supplied to each portion of the liquid ejecting apparatus **100**. In the example shown in FIG. **2**, the power supply circuit **113** generates a power supply potential VHV and an offset potential VBS. The offset potential VBS is supplied to the head chip **111** and the like. Further, the power supply potential VHV is supplied to the drive signal generation circuit **114** and the like.
(46) The drive signal generation circuit **114** is a circuit that generates a drive signal Com for driving each drive element **111***f* of the head chip **111**. Specifically, the drive signal generation circuit **114** includes, for example, a digital-to-analog (DA) conversion circuit and an amplifier circuit. The drive signal generation circuit **114** generates the drive signal Com by the DA conversion circuit converting a waveform designation signal dCom from the processing circuit **150**, which will be described later, from a digital signal to an analog signal, and the amplifier circuit amplifying the analog signal using the power supply potential VHV from the power supply circuit **113**. Here, among the waveforms included in the drive signal Com, the signal of the waveform actually supplied to the drive element **111***f* is the drive pulse PD.
(47) The storage circuit **116** is a device that stores various programs executed by the processing circuit **117** and various data processed by the processing circuit **117**. The storage circuit **116** includes, for example, a semiconductor memory.
(48) The storage circuit **116** stores a program PG**0** for ejecting ink. The program PG**0** is, for example, a program such as firmware that executes processing necessary for various operations of the control module **110***b* described above. The program PG**0** may be provided in a storage circuit **140** or a storage circuit **240**, which will be described later, instead of the storage circuit **116**.
(49) The processing circuit **117** is a device having a function of controlling each portion of a control module **110***c* and a function of processing various data. The processing circuit **117** has one

or more processors such as, for example, a CPU. The processing circuit **117** may be integrally formed with the storage circuit **116**, and may be constituted by hardware such as a DSP, an ASIC, a PLD, or an FPGA.
(50) The processing circuit **117** controls each portion of the control module **110***c* by reading the program PG**0** from the storage circuit **116** and executing the program PG**0**.
(51) The moving mechanism **120** changes a relative position between the head unit **110** and the print medium. More specifically, when the liquid ejecting apparatus **100** is a serial type, the moving mechanism **120** includes a transport mechanism for transporting the print medium in a predetermined direction and a moving mechanism for repeatedly moving the head unit **110** along an axis orthogonal to the transport direction of the print medium. Further, when the liquid ejecting apparatus **100** is a line type, the moving mechanism **120** includes a transport mechanism for transporting the print medium in a direction intersecting a longitudinal direction of the elongated head unit **110**.
(52) The communication device **130** is a circuit capable of communicating with the first processing apparatus **200**. For example, the communication device **130** is an interface such as a wireless or wired local area network (LAN) or a universal serial bus (USB). USB is a registered trademark. The communication device **130** may be connected to another first processing apparatus **200** through another network such as the Internet. Further, the communication device **130** may be integrated with the processing circuit **150**.
(53) The storage circuit **140** stores various programs executed by the processing circuit **150** and various data, such as the recorded data DP, processed by the processing circuit **150**. The storage circuit **140** may include, for example, one or both semiconductor memories of one or more volatile memories such as random access memory (RAM) and one or more non-volatile memories such as read only memory (ROM), electrically erasable programmable read-only memory (EEPROM) or programmable ROM (PROM). The recorded data DP is supplied from, for example, the first processing apparatus **200**. The storage circuit **140** may be built as a part of the processing circuit **150**.
(54) The processing circuit **150** has a function of controlling the operation of each portion of the liquid ejecting apparatus **100** and a function of processing various data. The processing circuit **150** includes, for example, one or more processors such as a central processing unit (CPU). The processing circuit **150** may include a programmable logic device such as a field-programmable gate array (FPGA) in place of the CPU or in addition to the CPU.
(55) The processing circuit **150** controls the operation of each portion of the liquid ejecting apparatus **100** by executing a program stored in the storage circuit **140**. Here, the processing circuit **150** generates signals such as a control signal Sk, a print data signal SI, and the waveform designation signal dCom as signals for controlling the operation of each portion of the liquid ejecting apparatus **100**.
(56) The control signal Sk is a signal for controlling the driving of the moving mechanism **120** and is input to the moving mechanism **120**. The print data signal SI is a signal for controlling the drive of the drive circuit **112**, and is input to the drive circuit **112**. Specifically, the print data signal SI specifies whether the drive circuit **112** supplies the drive signal Com from the drive signal generation circuit **114** to the drive element **111***f* as a drive pulse PD, for each predetermined unit period. By the specifying, the amount of ink ejected from the head chip **111** and the like are specified. The waveform designation signal dCom is a digital signal for defining the waveform of the drive signal Com generated by the drive signal generation circuit **114**, and is input to the drive signal generation circuit **114**.
(57) FIG. **3** is a cross-sectional view showing a configuration example of the head chip **111**. In the following description, for convenience of description, an X axis, a Y axis and a Z axis that intersect each other are appropriately used. In the following, one direction along the X axis is an X**1** direction, and a direction opposite to the X**1** direction is an X**2** direction. Similarly, the directions

opposite to each other along the Y axis are a Y**1** direction and a Y**2** direction. Opposite directions along the Z axis are a Z**1** direction and a Z**2** direction. The configuration of the head chip **111** is an example and is not limited to the example shown in FIG. **3**.
(58) As shown in FIG. **3**, the head chip **111** has a plurality of nozzles N arranged in a direction along the Y axis. The plurality of nozzles N are divided into a first column L**1** and a second column L**2** which are arranged at intervals in a direction along the X axis. Each of the first column L**1** and the second column L**2** is a set of a plurality of nozzles N linearly arranged in the direction along the Y axis.
(59) The head chip **111** has a configuration substantially symmetrical with each other in the direction along the X axis. However, positions of the plurality of nozzles N in the first column L**1** and the plurality of nozzles N in the second column L**2** in the direction along the Y axis may match or differ from each other. FIG. **3** illustrates a configuration in which the positions of the plurality of nozzles N in the first column L**1** and the plurality of nozzles N in the second column L**2** in the direction along the Y axis match with each other.
(60) As shown in FIG. **3**, the head chip **111** includes a flow path substrates **111***a*, a pressure chamber substrate **111***b*, a nozzle plate **111***c*, vibration absorbing bodies **111***d*, a vibration plate **111***e*, a plurality of drive elements **111***f*, protective plates **111***g*, a case **111***h*, and a wiring substrate **111***i*.
(61) The flow path substrate **111***a* and the pressure chamber substrate **111***b* are stacked in this order in the Z**1** direction, and form a flow path for supplying ink to a plurality of nozzles N. The vibration plate **111***e*, the plurality of drive elements **111***f*, the protective plates **111***g*, the case **111***h*, and the wiring substrate **111***i* are installed in a region positioned in the Z**1** direction with respect to the stack body formed by the flow path substrate **111***a* and the pressure chamber substrate **111***b*. On the other hand, the nozzle plate **111***c* and the vibration absorbing bodies **111***d* are installed in a region positioned in the Z**2** direction with respect to the stack body. Each element of the head chip **111** is schematically a plate-shaped member elongated in the Y direction, and is joined to each other by, for example, an adhesive. Hereinafter, each element of the head chip **111** will be described in order.
(62) The nozzle plate **111***c* is a plate-shaped member provided with a plurality of nozzles N in each of the first column L**1** and the second column L**2**. Each of the plurality of nozzles N is a through hole through which ink is passed. Here, the surface of the nozzle plate **111***c* facing the Z**2** direction is a nozzle surface FN. The nozzle plate **111***c* is manufactured by processing a silicon single crystal substrate by a semiconductor manufacturing technique using a processing technique such as dry etching or wet etching, for example. However, other known methods and materials may be appropriately used for manufacturing the nozzle plate **111***c*. Further, the cross-sectional shape of the nozzle is typically a circular shape, but the shape is not limited thereto, and may be a non-circular shape such as a polygon or an ellipse.
(63) The flow path substrate **111***a* is provided with a space R**1**, a plurality of supply flow paths Ra, and a plurality of communication flow paths Na for each of the first column L**1** and the second column L**2**. The space R**1** is an elongated opening extending in the direction along the Y axis in a plan view in the direction along the Z axis. Each of the supply flow path Ra and the communication flow path Na is a through hole formed for each nozzle N. Each supply flow path Ra communicates with the space R**1**.
(64) The pressure chamber substrate **111***b* is a plate-shaped member provided with a plurality of pressure chambers C referred to as cavities for each of the first column L**1** and the second column L**2**. The plurality of pressure chambers C are arranged in the direction along the Y axis. Each pressure chamber C is an elongated space formed for each nozzle N and extending in the direction along the X axis in a plan view. Each of the flow path substrate **111***a* and the pressure chamber substrate **111***b* is manufactured by processing a silicon single crystal substrate by, for example, semiconductor manufacturing technique, in the same manner as the nozzle plate **111***c* described

above. However, other known methods and materials may be appropriately used for the manufacturing of each of the flow path substrate **111***a* and the pressure chamber substrate **111***b*.
(65) The pressure chamber C is a space positioned between the flow path substrate **111***a* and the vibration plate **111***e*. For each of the first column L**1** and the second column L**2**, a plurality of the pressure chambers C are arranged in a direction along the Y axis. Further, the pressure chamber C communicates with each of the communication flow path Na and the supply flow path Ra. Therefore, the pressure chamber C communicates with the nozzle N through the communication flow path Na and communicates with the space R**1** through the supply flow path Ra.
(66) The vibration plate **111***e* is arranged on the surface of the pressure chamber substrate **111***b* facing the Z**1** direction. The vibration plate **111***e* is a plate-shaped member that can elastically vibrate. The vibration plate **111***e* has, for example, a first layer and a second layer, which are stacked in the Z**1** direction in this order. The first layer is, for example, an elastic film made of silicon oxide (SiO.sub.2). The elastic film is formed, for example, by thermally oxidizing one surface of a silicon single crystal substrate. The second layer is, for example, an insulating film made of zirconium oxide (ZrO.sub.2). The insulating film is formed by, for example, forming a zirconium layer by a sputtering method and thermally oxidizing the layer. The vibration plate **111***e* is not limited to the above-mentioned stacked configuration of the first layer and the second layer, and may be constituted by, for example, a single layer or three or more layers.
(67) On the surface of the vibration plate **111***e* facing the Z**1** direction, a plurality of drive elements **111***f* corresponding to the nozzles N are arranged for each of the first column L**1** and the second column L**2**. Each drive element **111***f* is a passive element that is deformed by the supply of the drive signal. Each drive element **111***f* has an elongated shape extending in the direction along the X axis in a plan view. The plurality of drive elements **111***f* are arranged in the direction along the Y axis to correspond to the plurality of pressure chambers C. The drive element **111***f* overlaps the pressure chamber C in a plan view.
(68) Each drive element **111***f* is a piezoelectric element, and although not shown, it has a first electrode, a piezoelectric layer, and a second electrode, which are stacked in the Z**1** direction in this order. One of the first electrode and the second electrode is an individual electrode arranged apart from the other for each drive element **111***f*, and the drive pulse PD is supplied to the one electrode. The other electrode of the first electrode and the second electrode is a band-shaped common electrode extending in the direction along the Y axis to be continuous over the plurality of drive elements **111***f*, and the offset potential VBS is supplied to the other electrode. Examples of the metal material of the electrodes include metal materials such as platinum (Pt), aluminum (Al), nickel (Ni), gold (Au), and copper (Cu), and of the materials, one type can be used alone or two or more types can be used in combination in an alloyed or stacked manner. The piezoelectric layer is made of a piezoelectric material such as lead zirconate titanate (Pb(Zr, Ti) O.sub.3), and has, for example, a band shape extending in the direction along the Y axis to be continuous over the plurality of drive elements **111***f*. However, the piezoelectric layer may be integrated over the plurality of drive elements **111***f*. In this case, the piezoelectric layer is provided with a through hole penetrating the piezoelectric layer extending in the direction along the X axis in a region corresponding to the gap between the pressure chambers C adjacent to each other in a plan view. When the vibration plate **111***e* vibrates in conjunction with the above deformation of the drive elements **111***f*, the pressures in the pressure chambers C fluctuate, and ink is ejected from the nozzles N.
(69) The protective plates **111***g* are a plate-shaped members installed on the surface of the vibration plate **111***e* facing the Z**1** direction, and protect the plurality of drive elements **111***f* and reinforce the mechanical strength of the vibration plate **111***e*. Here, the plurality of drive elements **111***f* are accommodated between the protective plates **111***g* and the vibration plate **111***e*. The protective plates **111***g* are made of, for example, a resin material.
(70) The case **111***h* is a member for storing ink supplied to a plurality of pressure chambers C. The

case **111***h* is made of, for example, a resin material. The case **111***h* is provided with a space R**2** for each of the first column L**1** and the second column L**2**. The space R**2** is a space communicating with the above-mentioned space R**1** and functions as a reservoir R for storing ink supplied to a plurality of pressure chambers C together with the space R**1**. The case **111***h* is provided with an introduction port IH for supplying ink to each reservoir R. The ink in each reservoir R is supplied to the pressure chamber C through each supply flow path Ra.
(71) The vibration absorbing body **111***d*, also referred to as a compliance substrate, is a flexible resin film constituting a wall surface of the reservoir R, and absorbs pressure fluctuations of ink in the reservoir R. The vibration absorbing body **111***d* may be a thin plate made of metal and having flexibility. The surface of the vibration absorbing body **111***d* facing the Z**1** direction is joined to the flow path substrate **111***a* with an adhesive or the like.
(72) The wiring substrate **111***i* is mounted on the surface of the vibration plate **111***e* facing the Z**1** direction, and is a mounting component for electrically coupling the head chip **111**, the drive circuit **112**, the control module **110***b*, and the like. The wiring substrate **111***i* is a flexible wiring substrate such as a chip on film (COF), a flexible printed circuit (FPC) or a flexible flat cable (FFC). The drive circuit **112** described above is mounted on the wiring substrate **111***i* of the present embodiment.
(73) 1-3. Configuration of First Processing Apparatus
(74) FIG. **4** is a schematic diagram showing a configuration example of the first processing apparatus **200** used in the liquid ejecting system **10** according to the first embodiment. As shown in FIG. **4**, the first processing apparatus **200** includes a display device **210**, an input device **220**, a communication device **230**, a storage circuit **240**, and a processing circuit **250**. The components are communicably connected to each other.
(75) The display device **210** displays various images under the control of the processing circuit **250**. Here, the display device **210** includes a display region **211** constituted by a display panel such as a liquid crystal display panel or an organic electro-luminescence (EL) display panel. The display device **210** may be provided outside the first processing apparatus **200**. Further, the display device **210** may be a component of the liquid ejecting apparatus **100**.
(76) The input device **220** is a device that receives operations from the user. For example, the input device **220** has a pointing device such as a touch pad, a touch panel or a mouse. Here, when the input device **220** has a touch panel, the input device **220** may also serve as a display device **210**. The input device **220** may be provided outside the first processing apparatus **200**. Further, the input device **220** may be a component of the liquid ejecting apparatus **100**. Further, the input device **220** may include an image capturing device having a charge coupled device (CCD) image sensor, a complementary MOS (CMOS) image sensor, or the like.
(77) The communication device **230** is a circuit capable of communicating with each of the liquid ejecting apparatus **100** and the external server **300**. For example, the communication device **230** is an interface such as a wireless or wired LAN or USB. The communication device **230** transmits the recorded data DP to the liquid ejecting apparatus **100** by communicating with the liquid ejecting apparatus **100**. Further, the communication device **230** transmits the unique information D**1** and receives the update information D**2** by communicating with the external server **300**. That is, the communication device **230** functions as a first connection portion **231** that is communicably connected to the external server **300**. The communication device **230** may be integrated with the processing circuit **250**.
(78) The storage circuit **240** is a device that stores various programs executed by the processing circuit **250** and various data processed by the processing circuit **250**. The storage circuit **240** has, for example, a hard disk drive or a semiconductor memory. A part or all of the storage circuit **240** may be provided in a storage device or a server outside the first processing apparatus **200**.
(79) The storage circuit **240** of the present embodiment stores a program PG**1**, the unique information D**1**, the update information D**2**, arrangement information D**3**, and the recorded data

DP. Some or all of the program PG**1**, the unique information D**1**, the update information D**2**, and the arrangement information D**3** may be stored in a storage device or server outside the first processing apparatus **200**. Further, in the following, the program PG**1**, the unique information D**1**, and the update information D**2** may be collectively referred to as information DG.
(80) The program PG**1** is a management program for managing the liquid ejecting apparatus **100**, and causes a computer to implement various functions required for the management.
(81) The unique information D**1** is information unique to the head unit **110**. More specifically, the unique information D**1** may be any information that can identify the type of the head unit **110**, and examples thereof may include identification information indicating a unique number such as a serial number unique to the head unit **110** or the liquid ejecting head **110***a*.
(82) In addition, other information about the head unit **110** may be added to the unique information D**1**. Examples of the other information include information regarding the type of liquid used for ejection in the head unit **110**, information regarding the temperature of the head unit **110**, and the like. The Information regarding the type of liquid used for ejection in the head unit **110** includes, for example, information regarding a product number, viscosity, residual vibration, ejection speed, and the like, of the ink. Examples of the information regarding the temperature of the head unit **110** include information regarding the detection temperature of a temperature sensor provided around the head unit **110**.
(83) The update information D**2** is information related to the update of the head unit **110**. As described above, the update information D**2** is provided from the external server **300** to the first processing apparatus **200**. In the example shown in FIG. **4**, the update information D**2** includes deficiency notification information D**2***a* and update information D**2***b*.
(84) The deficiency notification information D**2***a* is information for notifying the user of the deficiency of the head unit **110**. More specifically, the deficiency notification information D**2***a* is, for example, information indicating the existence of a defect or the like discovered after manufacturing in the head unit **110** having a specific serial number (ID).
(85) The update information D**2***b* is information such as a program for updating the program PG**0** of the head unit **110** described above.
(86) The arrangement information D**3** is information regarding arrangement of a plurality of liquid ejecting heads **110***a*. More specifically, the arrangement information D**3** is, for example, information indicating arrangement of a plurality of liquid ejecting heads **110***a* for each ink color.
(87) The processing circuit **250** is a device having a function of controlling each portion of the first processing apparatus **200** and a function of processing various data. The processing circuit **250** has, for example, a processor such as a central processing unit (CPU). The processing circuit **250** may be constituted by a single processor or may be constituted by a plurality of processors. In addition, some or all of the functions of the processing circuit **250** are implemented by hardware such as a digital signal processor (DSP), an application specific integrated circuit (ASIC), a programmable logic device (PLD), a field programmable gate array (FPGA).
(88) The processing circuit **250** functions as an acquisition portion **251**, a first output portion **252**, a first input portion **253**, a notification portion **254**, a display control portion **255**, a first receiving portion **256**, and a second receiving portion **257** by reading the program PG**1** from the storage circuit **240** and executing the program PG**1**.
(89) The acquisition portion **251** acquires the unique information D**1**. In the present embodiment, the acquisition portion **251** acquires the unique information D**1** based on a reception result in the first receiving portion **256**. When the storage circuit **116** of the head unit **110** stores identification information indicating a unique number such as a serial number, the acquisition portion **251** may acquire the unique information D**1** by reading the identification information.
(90) The first output portion **252** outputs the unique information D**1** through the first connection portion **231**. For example, the first output portion **252** outputs the unique information D**1** to the external server **300** through the first connection portion **231**, by using an instruction or the like

issued by a user using the input device **220** as a trigger. The first output portion **252** may output other information such as the arrangement information D**3** together with the unique information D**1** to the external server **300**.
(91) The update information D**2** is input to the first input portion **253** through the first connection portion **231**. For example, the update information D**2** is input from the external server **300** to the first input portion **253** through the first connection portion **231**, by using an instruction or the like issued by the user using the input device **220** as a trigger.
(92) The notification portion **254** notifies the user based on the update information D**2**. In the present embodiment, the notification portion **254** makes the notification by using the display device **210**. More specifically, the notification portion **254** causes the display device **210** to display information for the notification in the display region **211**. A specific example of the notification will be described later with reference to FIG. **9**.
(93) The display control portion **255** displays the arrangement image GA showing the arrangement of the plurality of liquid ejecting heads **110***a* in the display region **211**. In the present embodiment, the display control portion **255** displays the arrangement image GA in the display region **211** based on the arrangement information D**3**. In addition, the display control portion **255** displays an appropriate image such as an error image B**2** in the display region **211** based on a reception result of the first receiving portion **256**. A specific example of the arrangement image GA will be described later with reference to FIGS. **7**, **8**, and **10** to **15**.
(94) The first receiving portion **256** receives, from the user, an operation on the arrangement image GA. Based on the result of the receiving, the display control portion **255** performs display control of the display region **211**. The second receiving portion **257** receives an input from the user regarding the arrangement information D**3**. The arrangement information D**3** is generated based on the result of the receiving. Details of the receiving portion will be described later with reference to FIG. **8**.
(95) 1-4. Configuration of Server
(96) FIG. **5** is a schematic diagram showing a configuration example of the external server **300** used in the liquid ejecting system **10** according to the first embodiment. As shown in FIG. **5**, the external server **300** includes a display device **310**, an input device **320**, a communication device **330**, a storage circuit **340**, and a processing circuit **350**. The components are communicably connected to each other.
(97) The display device **310** is a device that displays various images under the control of the processing circuit **350**, and is configured in the same manner as the display device **210** described above. The input device **320** is a device that receives an operation from the user, and is configured in the same manner as the input device **220** described above. The communication device **330** is a circuit capable of communicating with each first processing apparatus **200**, and is configured in the same manner as the communication device **230** as described above. The communication device **330** may be integrated with the processing circuit **350**.
(98) Here, the communication device **330** receives the unique information D**1** and transmits the update information D**2** by communicating with the first processing apparatus **200**. That is, the communication device **330** functions as a second connection portion **331** that is communicably connected to the first connection portion **231**. In addition, the communication device **330** transmits the unique information D**1** by communicating with the second processing apparatus **400**, as needed.
(99) The storage circuit **340** is a device that stores various programs executed by the processing circuit **350** and various data processed by the processing circuit **350**, and is configured in the same manner as the storage circuit **240** described above. The storage circuit **340** stores a program PG**2**, the unique information D**1**, the update information D**2**, and correspondence information D**4**. When the first processing apparatus **200** transmits the arrangement information D**3** together with the unique information D**1**, the arrangement information D**3** is also stored in the storage circuit **340**.
(100) The program PG**2** is a program that enables a computer to implement various functions

necessary for generating the update information D**2** according to the unique information D**1**. The correspondence information D**4** is information regarding a correspondence relationship between the unique information D**1** and the update information D**2**. For the correspondence information D**4**, any format is possible without being limited to a particular format; for example, the format is a lookup table showing the correspondence relationship between the unique information D**1** and the update information D**2**. Further, the "generation of the update information D**2**" includes a case where one or more pieces of update information D**2** are selected or extracted from a plurality of pieces of update information D**2**.

(101) For example, after the head manufacturer manufactures and ships the head unit **110**, in some cases, some deficiency may be found in the head unit **110** or the program PG**0** mounted on the head unit **110** may be updated. In this case, the head manufacturer wants to notify the users of the information related to update, such as the deficiency notification or updates, but when the printer manufacturer is involved in the sale, the head manufacturer may not be able to specify the users. Therefore, the head manufacturer stores, in the external server **300**, the correspondence relationship in which the update information D**2** and the unique information D**1** about the head unit corresponding to the update information D**2** are associated with the external server **300**, as the correspondence information D**4**. Then, when the unique information D**1** is input to the external server **300** from the first processing apparatus **200** on the user side, the external server **300** determines whether or not the unique information D**1** is included in the correspondence information D**4**. When the unique information D**1** is included in the correspondence information D**4**, the external server **300** outputs the update information D**2** corresponding to the unique information D**1** to the first processing apparatus **200** in the correspondence information D**4**. In this way, even if the user cannot be specified on the head manufacturer side, the user can be notified of the update information D**2** according to the unique information D**1**.

(102) The processing circuit **350** is a device having a function of controlling each portion of the external server **300** and a function of processing various data, and is configured in the same manner as the processing circuit **250** described above. The processing circuit **350** functions as an output portion **351**, an input portion **352**, and a calculation portion **353** by reading the program PG**2** from the storage circuit **340** and executing the program PG**2**.

(103) The output portion **351** outputs the update information D**2** through the second connection portion **331**. For example, the output portion **351** outputs the update information D**2** to the first processing apparatus **200** through the second connection portion **331**, by using an instruction or the like issued by the user using the input device **220** as a trigger.

(104) The unique information D**1** is input to the input portion **352** through the second connection portion **331**. For example, the unique information D**1** is input from the first processing apparatus **200** to the input portion **352** through the second connection portion **331**, by using an instruction or the like by the user using the input device **220** as a trigger.

(105) The calculation portion **353** performs a calculation to generate the update information D**2** according to the unique information D**1**. In the present embodiment, the calculation portion **353** performs a calculation to generate the update information D**2** using the unique information D**1** and the correspondence information D**4**.

(106) 1-5. Process of Liquid Ejecting System

(107) FIG. **6** is a flowchart showing a process of the liquid ejecting system **10** according to the first embodiment. In the liquid ejecting system **10**, first, as shown in FIG. **6**, in step S**101**, the first processing apparatus **200** acquires the unique information D**1**. This acquisition is performed by the above-mentioned acquisition portion **251**. Here, the acquisition portion **251** causes the display device **210** to display an input image GD, which will be described later, as an image for a graphical user interface (GUI) for inputting information required for acquiring the unique information D**1**. The first receiving portion **256** receives an operation on the input image GD from the user. The acquisition portion **251** acquires the unique information D**1** based on the result of the receiving by

the first receiving portion **256**.
(108) Then, in step S**102**, the first processing apparatus **200** outputs the unique information D**1** to the external server **300**.
(109) Specifically, in step S**102**, the first output portion **252** outputs the unique information D**1** through the first connection portion **231** by using the acquisition of the unique information D**1** as a trigger. A timing at which the unique information D**1** is output to the first connection portion **231** is not limited to the time when the unique information D**1** is acquired. For example, the first output portion **252** may transmit authentication information such as account information and a password for the user to the external server **300** when the first receiving portion **256** receives the input using an input image GD for the GUI to be described later, the external server **300** may transmit output permission information to the first processing apparatus **200** when the external server **300** succeeds in the authentication using the authentication information, and the first output portion **252** may output the unique information D**1** to the external server **300** when the output permission information is received by the first processing apparatus **200**.
(110) Next, in step S**103**, the external server **300** inputs the unique information D**1**. The input is performed by the input portion **352**. Then, in step S**104**, the external server **300** generates the update information D**2** based on the unique information D**1**. The generation is performed by the calculation portion **353**. The external server **300** may compare the unique information D**1** with predetermined comparison information, and may cancel the processing after step S**104** depending on the comparison result. In this case, the output portion **351** may output information indicating that the update information D**2** is not provided.
(111) Then, in step S**105**, the external server **300** outputs the update information D**2** to the first processing apparatus **200**. The output is performed by the output portion **351**.
(112) Next, in step S**106**, the first processing apparatus **200** inputs the update information D**2**. Specifically, in step S**106**, the update information D**2** from the external server **300** is input to the first input portion **253** through the first connection portion **231**. The first input portion **253** may notify the user as to whether or not the update information D**2** is input from the external server **300** by using the display device **210** or the like, and the update information D**2** from the external server **300** may be input only when the user inputs an instruction to permit an input by the input device **220** or the like.
(113) Then, in step S**107**, the first processing apparatus **200** makes a notification based on the update information D**2**. The notification is performed by the notification portion **254**. Specifically, the notification portion **254** displays a region RC**3** based on the update information D**2** in the display region **211**.
(114) 1-6. Display in Display Region
(115) When the above-described program PG**1** is executed by the first processing apparatus **200**, various information required to manage the liquid ejecting apparatus **100** is displayed in the display region **211** of the display device **210**. Hereinafter, the display in the display region **211** will be described by taking Arrangement Examples 1 to 4 of the plurality of liquid ejecting heads **110***a* as examples. It should be noted that Arrangement Examples 1 to 4 are examples for convenience in describing the arrangement of the plurality of liquid ejecting heads **110***a*, and the present disclosure is not limited thereto.
Arrangement Example 1
(116) FIG. **7** is a diagram showing Arrangement Example 1 of the plurality of liquid ejecting heads **110***a*. In FIG. **7**, an arrangement of four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** is shown as Arrangement Example 1. Each of the liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** is the liquid ejecting head **110***a* that ejects cyan ink.
(117) In Arrangement Example 1, as shown in FIG. **7**, the liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** are arranged along the same straight line in the Y**2** direction in this order.
(118) FIG. **8** is a diagram showing an example in which the arrangement image GA showing

Arrangement Example 1 is displayed in the display region **211**. In the example shown in FIG. **8**, a reception image GB is displayed in the display region **211** in addition to the arrangement image GA.
(119) The reception image GB is a GUI image for receiving an input of information required to generate the arrangement information D**3** by operating the input device **220** by the user. The second receiving portion **257** generates the arrangement information D**3** using the information input using the reception image GB.
(120) The reception image GB includes regions RB**1**, RB**2**, and RB**3**. The region RB**1** is a region for inputting the number of columns of the plurality of liquid ejecting heads **110***a* for each ink color. The region RB**2** is a region for inputting the number of rows of the plurality of liquid ejecting heads **110***a* for each ink color. The region RB**3** is a region for inputting the presence or absence of an offset. The offset will be described later with reference to FIG. **10**.
(121) In the example shown in FIG. **8**, each of the regions RB**1**, RB**2**, and RB**3** is a drop-down list. FIG. **8** shows a state in which information indicating Arrangement Example 1 is input to the regions RB**1**, RB**2**, and RB**3**. Here, in the region RB**1**, "1" indicating that the liquid ejecting heads **110***a* for cyan ink are in one column is input, and "Null" indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**2**, "4" indicating that the liquid ejecting heads **110***a* for cyan ink are in four rows is input, and "Null" indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**3**, "Null" indicating that the offset does not exist is input.
(122) For each of the regions RB**1**, RB**2**, and RB**3**, any region is possible as long as it can receive necessary information, and the mode is not limited to using the drop-down list and other known widgets can be applied.
(123) The arrangement image GA includes a plurality of head display regions B**1**. The plurality of head display regions B**1** are divided for each ink color.
(124) Specifically, the plurality of head display regions B**1** is divided into a cyan head region RA_C indicating the arrangement of a plurality of liquid ejecting heads **110***a* that use cyan ink, a magenta head region RA_M indicating the arrangement of a plurality of liquid ejecting heads **110***a* that use magenta ink, a yellow head region RA_Y indicating the arrangement of a plurality of liquid ejecting heads **110***a* that use yellow ink, and a black head region RA_B indicating the arrangement of a plurality of liquid ejecting heads **110***a* that uses black ink.
(125) In the example shown in FIG. **8**, each of the cyan head regions RA_C, the magenta head region RA_M, the yellow head region RA_Y, and the black head region RA_B includes eight head display regions B**1** having 4 rows and 2 columns. The number of rows and the number of columns of the head display regions B**1** constituting each of the cyan head region RA_C, the magenta head region RA_M, the yellow head region RA_Y, and the black head region RA_B are not limited to the example shown in FIG. **8**, and there may be any numbers of rows and columns.
(126) The plurality of head display regions B**1** have display modes varied depending on the arrangement of the plurality of liquid ejecting heads **110***a*. Here, the display mode of each of the plurality of head display regions B**1** can be varied depending on the color, the presence or absence of characters, and the like. In FIG. **8**, the cyan head region RA_C is displayed in a mode showing Arrangement Example 1. In the example shown in FIG. **8**, among the eight head display regions B**1** constituting the cyan head region RA_C, four head display regions B**1** according to Arrangement Example 1 are displayed in a manner different from the other four head display regions B**1**. More specifically, among the eight head display regions B**1** constituting the cyan head region RA_C, characters "C**1**", "C**2**", "C**3**", and "C**4**" are displayed in the head display regions B**1** corresponding to the arrangement of the liquid ejecting heads **110***a*.
(127) Here, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** arranged in the Y**1** direction are displayed side by side in a direction DR**1** corresponding to the Y**1** direction. As described above, by varying the position

and the number of the head display regions B**1** indicating the existence of the liquid ejecting heads **110***a* depending on the arrangement of the liquid ejecting heads **110***a*, the arrangement of the plurality of liquid ejecting heads **110***a* is displayed in the arrangement image GA.
(128) In the present embodiment, among the eight head display regions B**1** constituting the cyan head region RA_C, the head display region B**1** corresponding to the liquid ejecting head **110***a*_C**2** exhibits an error image B**2** indicating an error state. In the example shown in FIG. **8**, the error image B**2** is displayed in a color different from that of the other head display regions B**1**. The error image B**2** may vary in the display mode in, for example, a color and presence or absence of blinking, depending on the content of a property image GC and the like. The error image B**2** may be displayed in a region different from the arrangement image GA.
(129) The head display region B**1** can receive, from the user, an operation on the head display region B**1** indicating that the liquid ejecting head **110***a* exists among the plurality of head display regions B**1**. The operation is received by the first receiving portion **256**. When the first receiving portion **256** receives the operation, the display control portion **255** displays the property information about the target liquid ejecting head **110***a* in the display region **211**. An example of the display will be described with reference to FIG. **9**.
(130) FIG. **9** is a diagram showing an example in which the property image GC and the input image GD are displayed in the display region **211**. FIG. **9** shows a display of the display region **211** when, among the plurality of head display regions B**1**, the operation on the head display region B**1** corresponding to the liquid ejecting head **110***a*_C**2** is received.
(131) The input image GD is an image for a GUI for inputting information required to acquire the unique information D**1**. In the example illustrated in FIG. **9**, the input image GD has a button and a text box. Identification information indicating a unique number such as a serial number of the liquid ejecting head **110***a* is input to the text box. The button is operated after the identification information is input to the text box. In this way, the first processing apparatus **200** requests various information related to the unique information D**1** based on the identification information from the external server **300**. Then, the external server **300** transmits various information to the first processing apparatus **200**. The first processing apparatus **200** acquiring the various information displays the various information as the property image GC. The property image GC includes regions RC**1**, RC**2**, and RC**3**.
(132) Information about the expiration date of the liquid ejecting head **110***a* is displayed in the region RC**1**. In the example shown in FIG. **9**, in the region RC**1**, items such as a manufacturing date, a usage start date, a cumulative number of drives, and an estimated expiration date are displayed for the liquid ejecting head **110***a* to be operated with respect to the head display region B**1**. The expiration date is determined by the external server **300**, for example, based on one or both of the number of days elapsed from the usage start date, the cumulative number of drives, and the unique information D**1**. Here, when a certain period of time elapses from a head manufacturing date, the external server **300** may determine that the expiration date is reached regardless of the number of days elapsed from the usage start date and the cumulative number of drives. The item displayed in the region RC**1** is not limited to the example shown in FIG. **9**, as long as the user can be notified of the expiration date of the liquid ejecting head **110***a*.
(133) The region RC**2** indicates the quality of a temperature state, an energization state, and a flow path state of the liquid ejecting head **110***a* to be operated with respect to the head display region B**1**.
(134) In the region RC**3**, a notification from the head manufacturer is displayed based on the update information D**2**. For example, when the liquid ejecting head **110***a* to be operated with respect to the head display region B**1** is deficient in the region RC**3**, display is performed to that effect. Here, when the liquid ejecting head **110***a* to be operated for the head display region B**1** is deficient, for example, by displaying “notify” in the region RC**3**, a notification is issued to the effect that the use of the liquid ejecting head **110***a* to be operated with respect to the head display

region B**1** is not recommended. In this case, necessary information may be provided to the user later by the head manufacturer or the printer manufacturer in an appropriate method.

Arrangement Example 2

(135) FIG. **10** is a diagram showing Arrangement Example 2 of a plurality of liquid ejecting heads. In FIG. **10**, as Arrangement Example 2, an arrangement of four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** that eject cyan ink is shown.

(136) In Arrangement Example 2, as shown in FIG. **10**, the four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** are arranged in a staggered manner. Here, the four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** are aligned in this order in the Y**2** direction. However, the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3** are aligned along the same straight line along the Y axis so that the positions in the directions along the X axis match. The liquid ejecting heads **110***a*_C**2** and **110***a*_C**4** are positioned in the X**1** direction with respect to the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3**, and are aligned along the same straight line along the Y axis so that the positions in the directions along the X axis match.

(137) Further, ends of the liquid ejecting heads **110***a*_C**2** and **110***a*_C**4** in the X**2** direction are positioned further in the X**1** direction with respect to ends of the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3** in the X**1** direction. In addition, an end of the liquid ejecting head **110***a*_C**2** in the Y**1** direction is positioned further in the Y**1** direction with respect to an end of the liquid ejecting head **110***a*_C**1** in the Y**2** direction. Similarly, an end of the liquid ejecting head **110***a*_C**3** in the Y**1** direction is positioned further in the Y**1** direction with respect to an end of the liquid ejecting head **110***a*_C**2** in the Y**2** direction. An end of the liquid ejecting head **110***a*_C**4** in the Y**1** direction is positioned further in the Y**1** direction with respect to an end of the liquid ejecting head **110***a*_C**3** in the Y**2** direction. In this way, “offset” means that the ends of the plurality of liquid ejecting heads **110***a* arranged to be misaligned with each other in the directions along the X axis are arranged to overlap each other in the direction along the Y axis.

(138) FIG. **11** is a diagram showing an example in which the arrangement image GA showing Arrangement Example 2 is displayed in the display region **211**. The arrangement image GA and the reception image GB shown in FIG. **11** are the same as the arrangement image GA and the reception image GB shown in FIG. **8** above, except that they are applied to Arrangement Example 2.

(139) FIG. **11** shows a state in which information indicating Arrangement Example 2 is input to the regions RB**1**, RB**2**, and RB**3**. Here, in the region RB**1**, “1” indicating that the liquid ejecting heads **110***a* for cyan ink are in one column is input, and “Null” indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**2**, “4” indicating that the liquid ejecting heads **110***a* for cyan ink are in four rows is input, and “Null” indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**3**, an “Offset” indicating that an offset exists is input.

(140) In FIG. **11**, the cyan head region RA_C is displayed in a mode showing Arrangement Example 2. In the example shown in FIG. **11**, among the eight head display regions B**1** constituting the cyan head region RA_C, four head display regions B**1** according to Arrangement Example 2 are displayed in a manner different from the other four head display regions B**1**. More specifically, among the eight head display regions B**1** constituting the cyan head region RA_C, characters “C**1**”, “C**2**”, “C**3**”, and “C**4**” are displayed in the head display regions B**1** corresponding to the arrangement of the liquid ejecting heads **110***a*.

(141) Here, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3** arranged in the Y**1** direction are displayed side by side in the direction DR**1** corresponding to the Y**1** direction. Similarly, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**2** and **110***a*_C**4** arranged in the Y**1** direction are displayed side by side in the direction DR**1** corresponding to the Y**1** direction. In addition, among the liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4**, two head display regions B**1** corresponding to the two liquid ejecting heads **110***a* arranged to be adjacent to each other in the Y**1**

direction and to be misaligned with each other in the X**1** direction are displayed to be adjacent to each other in the direction DR**1** corresponding to the Y**1** direction and to be misaligned with each other in a direction DR**2** corresponding to the X**1** direction.

Arrangement Example 3

(142) FIG. **12** is a diagram showing Arrangement Example 3 of the plurality of liquid ejecting heads **110***a*. In FIG. **12**, as Arrangement Example 3, an arrangement of four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** that eject cyan ink is shown.

(143) In Arrangement Example 3, as shown in FIG. **12**, the four liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, **110***a*_C**3**, and **110***a*_C**4** are arranged in a matrix shape. Here, the liquid ejecting heads **110***a*_C**1** and **110***a*_C**2** are aligned along the same straight line along the Y axis so that the positions in the directions along the X axis match. The liquid ejecting heads **110***a*_C**3** and **110***a*_C**4** are positioned in the X**1** direction with respect to the liquid ejecting heads **110***a*_C**1** and **110***a*_C**2**, and are aligned along the same straight line along the Y axis so that the positions in the directions along the X axis match.

(144) Further, the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3** are arranged on the same straight line along the X axis so that both ends thereof in the directions along the Y axis are aligned with each other. Similarly, the liquid ejecting heads **110***a*_C**2** and **110***a*_C**4** are arranged on the same straight line along the X axis so that both ends thereof in the directions along the Y axis are aligned with each other.

(145) FIG. **13** is a diagram showing an example in which the arrangement image GA showing Arrangement Example 3 is displayed in the display region **211**. The arrangement image GA and the reception image GB shown in FIG. **13** are the same as the arrangement image GA and the reception image GB shown in FIG. **8** above, except that they are applied to Arrangement Example 3.

(146) FIG. **13** shows a state in which information indicating Arrangement Example 3 is input to the regions RB**1**, RB**2**, and RB**3**. Here, in the region RB**1**, “2” indicating that the liquid ejecting heads **110***a* for cyan ink are in two columns is input, and “Null” indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**2**, “2” indicating that the liquid ejecting heads **110***a* for cyan ink are in two rows is input, and “Null” indicating that liquid ejecting heads **110***a* for inks of other colors do not exist is input. In the region RB**3**, “Null” indicating that the offset does not exist is input.

(147) In FIG. **13**, the cyan head region RA_C is displayed in a mode showing Arrangement Example 3. In the example shown in FIG. **13**, among the eight head display regions B**1** constituting the cyan head region RA_C, four head display regions B**1** according to Arrangement Example 3 are displayed in a manner different from the other four head display regions B**1**. More specifically, among the eight head display regions B**1** constituting the cyan head region RA_C, characters “C**1**”, “C**2**”, “C**3**”, and “C**4**” are displayed in the head display regions B**1** corresponding to the arrangement of the liquid ejecting heads **110***a*.

(148) Here, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**1** and **110***a*_C**2** arranged in the Y**1** direction are displayed side by side in the direction DR**1** corresponding to the Y**1** direction. Similarly, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**3** and **110***a*_C**4** arranged in the Y**1** direction are displayed side by side in the direction DR**1** corresponding to the Y**1** direction. Further, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**1** and **110***a*_C**3** arranged in the X**1** direction are displayed side by side in the direction DR**2** corresponding to the X**1** direction. Similarly, the plurality of head display regions B**1** corresponding to the liquid ejecting heads **110***a*_C**2** and **110***a*_C**4** arranged in the X**1** direction are displayed side by side in the direction DR**2** corresponding to the X**1** direction.

Arrangement Example 4

(149) FIG. **14** is a diagram showing Arrangement Example 4 of the plurality of liquid ejecting heads **110***a*. In FIG. **14**, as Arrangement Example 4, an arrangement of three liquid ejecting heads

110*a*_C**1**, **110***a*_C**2**, and **110***a*_C**3**, three liquid ejecting heads **110***a*_M**1**, **110***a*_M**2**, and **110***a*_M**3**, and three liquid ejecting heads **110***a*_Y**1**, **110***a*_Y**2**, and **110***a*_Y**3** is shown. Each of the three liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, and **110***a*_C**3** is the liquid ejecting head **110***a* that ejects cyan ink. Each of the three liquid ejecting heads **110***a*_M**1**, **110***a*_M**2**, and **110***a*_M**3** is the liquid ejecting head **110***a* that ejects magenta ink. Each of the three liquid ejecting heads **110***a*_Y**1**, **110***a*_Y**2**, and **110***a*_Y**3** is the liquid ejecting head **110***a* that ejects yellow ink.
(150) In Arrangement Example 4, as shown in FIG. **14**, the three liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, and **110***a*_C**3** are arranged in a staggered manner, the three liquid ejecting heads **110***a*_M**1**, **110***a*_M**2**, and **110***a*_M**3** are arranged in a staggered manner, and the three liquid ejecting heads **110***a*_Y**1**, **110***a*_Y**2**, and **110***a*_Y**3** are arranged in a staggered manner.
(151) Here, the three liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, and **110***a*_C**3** are arranged in the same manner as the aforementioned Arrangement Example 2 shown in FIG. **10** except that the liquid ejecting heads **110***a*_C**4** are omitted. Each of the arrangement of the three liquid ejecting heads **110***a*_M**1**, **110***a*_M**2**, and **110***a*_M**3** and the arrangement of the three liquid ejecting heads **110***a*_Y**1**, **110***a*_Y**2**, and **110***a*_Y**3** is the same as the arrangement of the three liquid ejecting heads **110***a*_C**1**, **110***a*_C**2**, and **110***a*_C**3**.
(152) FIG. **15** is a diagram showing an example in which the arrangement image GA showing Arrangement Example 4 is displayed in the display region **211**. The arrangement image GA and the reception image GB shown in FIG. **15** are the same as the arrangement image GA and the reception image GB shown in FIG. **8** above, except that they are applied to Arrangement Example 4.
(153) FIG. **15** shows a state in which information indicating Arrangement Example 4 is input to the regions RB**1**, RB**2**, and RB**3**. Here, in the region RB**1**, "1" indicating that the liquid ejecting heads **110***a* for each of cyan ink, magenta ink, and yellow ink are in one column is input, and "Null" indicating that liquid ejecting heads **110***a* for black ink do not exist is input. In the region RB**2**, "3" indicating that the liquid ejecting heads **110***a* for each of cyan ink, magenta ink, and yellow ink are in three rows is input, and "Null" indicating that liquid ejecting heads **110***a* for black ink do not exist is input. In the region RB**3**, an "Offset" indicating that an offset exists is input.
(154) In FIG. **15**, the cyan head region RA_C is displayed in a mode showing Arrangement Example 4. In the example shown in FIG. **15**, among the eight head display regions B**1** constituting the cyan head region RA_C, three head display regions B**1** according to Arrangement Example 4 are displayed in a manner different from the other five head display regions B**1**. More specifically, among the eight head display regions B**1** constituting the cyan head region RA_C, characters "C**1**", "C**2**", and "C**3**" are displayed in the head display regions B**1** corresponding to the arrangement of the liquid ejecting heads **110***a*. Similarly, among the eight head display regions B**1** constituting the magenta head region RA_M, three head display regions B**1** according to Arrangement Example 4 are displayed in a manner different from the other five head display regions B**1**. Among the eight head display regions B**1** constituting the yellow head region RA_Y, three head display regions B**1** according to Arrangement Example 4 are displayed in a manner different from the other five head display regions B**1**.
(155) As described above, the liquid ejecting system **10** includes the head units **110**, the first output portion **252**, the first input portion **253**, and the notification portion **254**. The head units **110** eject ink, which is an example of "liquid". The first output portion **252** outputs the unique information D**1** unique to head unit **110** to the external server **300**. The update information D**2** for the head unit **110** is input from the external server **300** to the first input portion **253** in association with the unique information D**1**. The notification portion **254** notifies the user based on the update information D**2**.
(156) In the liquid ejecting system **10**, the first output portion **252** outputs the unique information D**1** to the external server **300**, and thus the external server **300** can generate the update information D**2** based on the unique information D**1**. The update information D**2** is input from the external server **300** to the first input portion **253**, the notification portion **254** performs the notification

based on the update information D**2**, and thus the update information D**2** is provided to the user. As described above, even when the printer manufacturer is different from the head manufacturer, the update information D**2** about the head unit **110** can be promptly provided to the user.
(157) As described above, the update information D**2** includes the deficiency notification information D**2***a* for notifying the user of the deficiency of the head unit **110**. Therefore, the user can be promptly notified of the deficiency of the head unit **110**.
(158) Further, as described above, the head unit **110** includes the storage circuit **116**, which is an example of a “storage portion”. The storage circuit **116** stores the program PG**0** used for ejecting ink. The update information D**2** includes the update information D**2***b* for updating the program PG**0**. Therefore, the update information D**2***b* can be promptly provided to the user.
(159) In the present embodiment, as described above, the liquid ejecting system **10** includes the liquid ejecting apparatus **100** including head units **110**, the first processing apparatus **200** that generates the recorded data DP used in the liquid ejecting apparatus **100**, and the external server **300**.
(160) Here, as described above, each of the first output portion **252**, the first input portion **253**, and notification portion **254** is provided in first processing apparatus **200**. Therefore, the first processing apparatus **200** can perform the display based on the update information D**2**.
(161) Further, as described above, the external server **300** stores correspondence information D**4** regarding the correspondence relationship between the unique information D**1** and the update information D**2**. Therefore, the external server **300** can generate the update information D**2** using the correspondence information D**4**.
2. Second Embodiment
(162) Hereinafter, a second embodiment of the present disclosure will be described. In the embodiment illustrated below, elements whose actions or functions are similar to those of the first embodiment will be denoted by the same reference numerals used in the description of the first embodiment and detailed description thereof will be omitted as appropriate.
(163) FIG. **16** is a schematic diagram showing a configuration example of a first processing apparatus **200**A in the second embodiment. The present embodiment is the same as the above-described first embodiment except that the first processing apparatus **200**A is used instead of the first processing apparatus **200**. The first processing apparatus **200**A has the same configuration as the first processing apparatus **200** except that a program PG**3** is used instead of the program PG**1** and an image capturing device **260** is added. The program PG**3** is the same as the program PG**1** except that the program PG**3** can cause a computer to execute a process of acquiring the unique information D**1** using the image capturing device **260**.
(164) The image capturing device **260** is a device having an image capturing element such as a CCD image sensor or a CMOS image sensor that captures an image of a plurality of liquid ejecting heads **110***a*. For example, the image capturing device **260** is disposed at a position facing the plurality of liquid ejecting heads **110***a*. The image capturing device **260** generates image capturing information D**5** indicating images obtained by imaging the plurality of liquid ejecting heads **110***a*. The image capturing information D**5** is stored in the storage circuit **240**. In addition to the image capturing element, the image capturing device **260** may include an optical system such as a lens, as needed.
(165) The processing circuit **250** functions as an acquisition portion **251**A, a first output portion **252**, a first input portion **253**, a notification portion **254**, a display control portion **255**, a first receiving portion **256**, and a second receiving portion **257** by reading the program PG**3** from the storage circuit **240** and executing the program PG**3**.
(166) The acquisition portion **251**A acquires the arrangement information D**3** in addition to the unique information D**1**. Here, the acquisition portion **251**A acquires the arrangement information D**3** based on the image capturing information D**5**. For example, the acquisition portion **251**A recognizes the arrangement of a plurality of liquid ejecting heads **110***a* by image processing for

performing object recognition on the liquid ejecting heads **110***a* from the image indicated by the image capturing information D**5**, and generates the arrangement information D**3** based on the recognition result.
(167) As in the above-described first embodiment, in the above-described second embodiment, the update information D**2** about the head unit **110** can be promptly provided to the user.
3. Third Embodiment
(168) Hereinafter, a third embodiment of the present disclosure will be described. In the embodiment illustrated below, elements whose actions or functions are similar to those of the first embodiment will be denoted by the same reference numerals used in the description of the first embodiment and detailed description thereof will be omitted as appropriate.
(169) FIG. **17** is a schematic diagram showing a configuration example of a liquid ejecting system **10**B according to the third embodiment. The liquid ejecting system **10**B has the same configuration as the liquid ejecting system **10** of the above-described first embodiment except that liquid ejecting apparatuses **100**B_**1** to **100**B_**3** and first processing apparatuses **200**B_**1** to **200**B_**3** are provided in place of the liquid ejecting apparatuses **100_1** to **100_3** and the first processing apparatuses **200_1** to **200_3**.
(170) The liquid ejecting apparatus **100**B_**1** is communicably connected to the first processing apparatus **200**B_**1** and is communicably connected to the external server **300** through the communication network NW. The liquid ejecting apparatus **100**B_**2** is communicably connected to the first processing apparatus **200**B_**2** and is communicably connected to the external server **300** through the communication network NW. The liquid ejecting apparatus **100**B_**3** is communicably connected to the first processing apparatus **200**B_**3** and is communicably connected to the external server **300** through the communication network NW. As described above, the liquid ejecting apparatuses **100**B_**1** to **100**B_**3** correspond to the first processing apparatuses **200**B_**1** to **200**B_**3**, respectively, and are communicably connected to the first processing apparatuses **200**B_**1** to **200**B_**3** and are communicably connected to the external server **300** through the communication network NW. In the following, without distinguishing each of the liquid ejecting apparatuses **100**B_**1** to **100**B_**3** from the others, they may be referred to as the liquid ejecting apparatus **100**B. Without distinguishing each of the first processing apparatuses **200**B_**1** to **200**B_**3** from the others, they may be referred to as the first processing apparatus **200**B.
(171) In the example shown in FIG. **17**, the number of each of the liquid ejecting apparatus **100**B and the first processing apparatus **200**B included in the liquid ejecting system **10**B is three, but the number is not limited thereto, and the number may be one, two, or four or more. That is, the number of the sets of the liquid ejecting apparatus **100**B and the first processing apparatus **200**B is not limited to three, and may be one, two, or four or more.
(172) The liquid ejecting apparatus **100**B is configured in the same manner as the liquid ejecting apparatus **100** of the first embodiment described above, except that a head unit **110**B is provided in place of the head unit **110**. The head unit **110**B is the same as the head unit **110** except that functions of acquiring and outputting the unique information D**1** and inputting and giving a notification of the update information D**2** are added. Details of the liquid ejecting apparatus **100**B will be described later with reference to FIG. **18**.
(173) The liquid ejecting apparatus **100**B outputs unique information D**1** to the external server **300** and inputs update information D**2** from the external server **300**. Then, the liquid ejecting apparatus **100**B makes the notification based on the update information D**2**.
(174) The first processing apparatus **200**B is configured in the same manner as the first processing apparatus **200** of the first embodiment described above, except that the functions of acquiring and outputting the unique information D**1** and inputting and giving the notification of the update information D**2** are omitted.
(175) FIG. **18** is a schematic diagram showing a configuration example of the liquid ejecting apparatus **100**B used in the liquid ejecting system **10**B according to the third embodiment. As

shown in FIG. **18**, the head unit **110**B included in the liquid ejecting apparatus **100**B is configured in the same manner as the head unit **110** of the first embodiment except that a control module **110***c* is provided in place of the control module **110***b*. The control module **110***c* is configured in the same manner as the control module **110***b* except that the information DG is used and a communication device **115** is added.
(176) The communication device **115** is a circuit capable of communicating with the external server **300**. For example, the communication device **115** is an interface such as a wireless or wired LAN or USB. The communication device **115** transmits the unique information D**1** and receives the update information D**2** by communicating with the external server **300**. That is, the communication device **115** functions as a first connection portion **115***a* that is communicably connected to the external server **300**, as similar to the first connection portion **231** of the first embodiment.
(177) The storage circuit **116** stores the same information DG as in FIG. **4** described above. That is, the storage circuit **116** stores the program PG**1**, the unique information D**1**, and the update information D**2**. The processing circuit **117** functions as an acquisition portion **117***a*, a first output portion **117***b*, a first input portion **117***c*, and a notification portion **117***d* by reading the program PG**1** from the storage circuit **116** and executing the program PG**1**.
(178) The acquisition portion **117***a* acquires the unique information D**1**, as similar to the acquisition portion **251** of the first embodiment. The first output portion **117***b* outputs the unique information D**1** through the first connection portion **115***a*, as similar to the first output portion **252** of the first embodiment. The update information D**2** is input to the first input portion **117***c* through the first connection portion **115***a*, as similar to the first input portion **253** of the first embodiment. The notification portion **117***d* makes a notification based on the update information D**2** as similar to the notification portion **254** of the first embodiment. The notification may be made by the display on the display region **211** of the first processing apparatus **200**B, by the display on the display device provided in the liquid ejecting apparatus **100**B, and so on.
(179) FIG. **19** is a flowchart showing a process of the liquid ejecting system **10**B according to the third embodiment. In the liquid ejecting system **10**B, first, as shown in FIG. **19**, in step S**201**, the liquid ejecting apparatus **100**B acquires the unique information D**1**. Then, in step S**202**, the liquid ejecting apparatus **100**B outputs the unique information D**1** to the external server **300**.
(180) Next, in step S**203**, the external server **300** inputs the unique information D**1**. Then, in step S**204**, the external server **300** generates the update information D**2** based on the unique information D**1**. Then, in step S**205**, the external server **300** outputs the update information D**2** to the liquid ejecting apparatus **100**B.
(181) Next, in step S**206**, the liquid ejecting apparatus **100**B inputs the update information D**2**. Then, in step S**207**, the liquid ejecting apparatus **100**B makes a notification based on the update information D**2**.
(182) As in the above-described first embodiment, in the above-described third embodiment, the update information D**2** about the head unit **110** can be promptly provided to the user. In the present embodiment, as described above, each of the first output portion **117***b*, the first input portion **117***c*, and the notification portion **117***d* is provided in the liquid ejecting apparatus **100**B. Therefore, the liquid ejecting apparatus **100**B can make the notification based on the update information D**2**. In addition, the program for notifying is not needed to be incorporated into the first processing apparatus **200**B, and thus it is possible to reduce the burden on the printer manufacturer or the user.
4. Fourth Embodiment
(183) Hereinafter, a fourth embodiment of the present disclosure will be described. In the embodiment illustrated below, elements whose actions or functions are similar to those of the first embodiment will be denoted by the same reference numerals used in the description of the first embodiment and detailed description thereof will be omitted as appropriate.
(184) FIG. **20** is a schematic diagram showing a configuration example of a liquid ejecting system **10**C according to the fourth embodiment. The liquid ejecting system **10**C has the same

configuration as that of the liquid ejecting system **10** of the first embodiment described above, except that an external server **300**C is provided instead of the external server **300** and a second processing apparatus **400**C is provided instead of the second processing apparatus **400**.
(185) The external server **300**C has the same configuration as that of the external server **300** of the first embodiment except that the external server **300**C outputs the unique information D**1** to the second processing apparatus **400**C, receives the input of the update information D**2** from the second processing apparatus **400**C, and outputs the update information D**2** to the first processing apparatus **200**.
(186) The second processing apparatus **400**C may have the same configuration as that of the second processing apparatus **400** of the first embodiment except that the second processing apparatus **400**C includes a second input portion **410** and a second output portion **420** and is capable of generating or inputting the update information D**2**. The second input portion **410** and the second output portion **420** are implemented by executing a program installed in the second processing apparatus **400**C on a processor of the second processing apparatus **400**C.
(187) The unique information D**1** is input from the external server **300**C to the second input portion **410**. The second output portion **420** outputs the update information D**2** to the external server **300**C. The update information D**2** to be output is generated in the same manner as in the external server **300** of the first embodiment described above, or is acquired by an operation of an operator of the second processing apparatus **400**C based on the input unique information D**1**.
(188) FIG. **21** is a flowchart showing a process of the liquid ejecting system **10**C according to the fourth embodiment. In the liquid ejecting system **10**C, first, as shown in FIG. **21**, in step S**301**, the first processing apparatus **200** acquires the unique information D**1**. Then, in step S**302**, the first processing apparatus **200** outputs the unique information D**1** to the external server **300**C.
(189) Next, in step S**303**, the external server **300** inputs the unique information D**1**. Then, in step S**304**, the external server **300** outputs the unique information D**1** to the second processing apparatus **400**C.
(190) Then, in step S**305**, the second processing apparatus **400**C inputs the unique information D**1**. Then, in step S**306**, the second processing apparatus **400**C acquires the update information D**2**. Then, in step S**307**, the second processing apparatus **400**C outputs the update information D**2** to the external server **300**C.
(191) Next, in step S**308**, the external server **300** inputs the update information D**2**. Then, in step S**309**, the external server **300** outputs the update information D**2** to the first processing apparatus **200**.
(192) Next, in step S**310**, the first processing apparatus **200** inputs the update information D**2**. Then, in step S**311**, the first processing apparatus **200** makes a notification based on the update information D**2**.
(193) As in the above-described first embodiment, in the above-described third embodiment, the update information D**2** about the head unit **110** can be promptly provided to the user. As described above, in the present embodiment, the liquid ejecting system **10**C further includes a second processing apparatus **400**C different from the first processing apparatus **200**. The second processing apparatus **400**C includes the second input portion **410** and the second output portion **420**. The unique information D**1** is input from the external server **300** to the second input portion **410**. The second output portion **420** outputs the update information D**2** to the external server **300**. Therefore, it is possible to generate the update information D**2**, or to generate and update the correspondence information D**4** by the second processing apparatus **400**C.
5. Modification Examples
(194) The liquid ejecting system of the present disclosure has been described above based on the illustrated embodiments, but the present disclosure is not limited thereto. Further, the configuration of each portion of the present disclosure can be replaced with any configuration that exhibits the same functions as that of the above-described embodiments, or any configuration can be added.

5-1. Modification Example 1
(195) In the above-described embodiments, the configuration in which the update information D**2** includes both the deficiency notification information D**2***a* and the update information D**2***b* is exemplified; however, in the update information D**2**, one of the deficiency notification information D**2***a* and the update information D**2***b* may be omitted, or information other than the deficiency notification information D**2***a* and the update information D**2***b* may be included.
5-2. Modification Example 2
(196) In the above-described embodiments, the configuration in which the external server **300** is a cloud server is exemplified, but the configuration is not limited thereto. For example, the external server **300** may be a server other than a cloud server or a virtual server, or may be an on-premises server.
5-3. Modification Example 3
(197) In the above-described embodiments, the configuration in which the drive element **111***f* is a piezoelectric element is exemplified, but the configuration is not limited thereto, and for example, the drive element **111***f* may be a heater that heats ink in the pressure chamber C. That is, the drive method of the head chip **111** is not limited to the piezoelectric method, and may be, for example, a thermal method.

Claims

1. A liquid ejecting system comprising: a liquid ejecting apparatus; a head unit that ejects liquid included in the liquid ejecting apparatus; a first processing apparatus that generates recorded data used in the liquid ejecting apparatus; a second processing apparatus that is different from the first processing apparatus; an external server; a first output portion that outputs unique information unique to the head unit to the external server; a first input portion to which update information about the head unit is input from the external server in association with the unique information; and a notification portion that notifies a user based on the update information, wherein the second processing apparatus includes a second input portion to which the unique information is input from the external server, and a second output portion that outputs the update information to the external server.
2. The liquid ejecting system according to claim 1, wherein the update information includes deficiency notification information for notifying the user of a deficiency of the head unit.
3. The liquid ejecting system according to claim 1, wherein the head unit includes a storage portion that stores a program used for ejecting liquid, and the update information includes update information for updating the program.
4. The liquid ejecting system according to claim 1, wherein each of the first output portion, the first input portion, and the first notification portion is provided in the liquid ejecting apparatus or the first processing apparatus.
5. The liquid ejecting system according to claim 1, wherein the external server stores correspondence information regarding a correspondence relationship between the unique information and the update information.